Students will be assessed on their ability to:		Suggested experiments
38	discuss situations that require the accurate determination of refractive index	
39	investigate and explain what is meant by *plane polarised light*	Models of structures to investigate stress concentrations
40	investigate and explain how to measure the rotation of the plane of polarisation	
41	investigate and recall that waves can be diffracted and that substantial diffraction occurs when the size of the gap or obstacle is similar to the wavelength of the wave	Demonstration using a ripple tank
42	explain how diffraction experiments provide evidence for the wave nature of electrons	
43	discuss how scientific ideas may change over time, for example, our ideas on the particle/wave nature of electrons	
44	recall that, in general, waves are transmitted and reflected at an interface between media	Demonstration using a laser
45	explain how different media affect the transmission/reflection of waves travelling from one medium to another	
46	explore and explain how a pulse-echo technique can provide details of the position and/or speed of an object and describe applications that use this technique	
47	explain qualitatively how the movement of a source of sound or light relative to an observer/detector gives rise to a shift in frequency (Doppler effect) and explore applications that use this effect	Demonstration using a ripple tank or computer simulation
48	explain how the amount of detail in a scan may be limited by the wavelength of the radiation or by the duration of pulses	
49	discuss the social and ethical issues that need to be considered, e.g. when developing and trialing new medical techniques on patients or when funding a space mission	

2.4 DC Electricity

This topic covers the definitions of various electrical quantities, for example, current and resistance, Ohm's law and non-ohmic materials, potential dividers, emf and internal resistance of cells, and negative temperature coefficient thermistors.

This topic may be studied using applications that relate to, for example, technology in space.

Students will be assessed on their ability to:		Suggested experiments
50	describe electric current as the rate of flow of charged particles and use the expression $I = \Delta Q/\Delta t$	
51	use the expression $V = W/Q$	
52	recognise, investigate and use the relationships between current, voltage and resistance for series and parallel circuits, and know that these relationships are a consequence of the conservation of charge and energy	Measure current and voltage in series and parallel circuits Use an ohmmeter to measure total resistance of series/parallel circuits
53	investigate and use the expressions $P = VI$, $W = VIt$. Recognise and use related expressions, e.g. $P = I^2R$ and $P = V^2/R$	Measure the efficiency of an electric motor (see 17)
54	use the fact that resistance is defined by $R = V/I$ and that Ohm's law is a special case when $I \alpha V$	
55	demonstrate an understanding of how ICT may be used to obtain current-potential difference graphs, including non-ohmic materials and compare this with traditional techniques in terms of reliability and validity of data	
56	interpret current-potential difference graphs, including non-ohmic materials	Investigate I–V graphs for filament lamp, diode and thermistor
57	investigate and use the relationship $R = \rho l/A$	Measure resisitivity of a metal and polythene
58	investigate and explain how the potential along a uniform current-carrying wire varies with the distance along it and how this variation can be made use of in a potential divider	Use a digital voltmeter to investigate 'output' of a potential divider